

Contact

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au

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Top Skills

k-means clustering
Q Learning
Collaborative Filtering

Certifications

EE585x: Space Mission Design and
Operations
Google Cloud Certified Professional
Machine Learning Engineer
Sub 7-kg Remote Pilot License
(RePL)
Neural Networks and Deep Learning
Structuring Machine Learning
Projects

Honors-Awards

O.H. Atkinson Cup - Combined Sport
& Study Award
SACE Certificate of Merit (Physics)
SACE Certificate of Merit
(Chemistry)
SACE Certificate of Merit (Business
& Enterprise)
Certificate of Merit (Attainment)

Patrick Capaldo

AI Engineer @ Airbus # | Google Certified Professional Cloud
Architect and Machine Learning Engineer | Volunteer @ IET
Bristol, England, United Kingdom

Summary

My dream is to pursue a career at the forefront of space exploration, aerospace engineering, and autonomy by contributing to the development of revolutionary design and technology. As an AI Engineer at Airbus, I am working towards this dream.

Experience

Airbus

2 years 1 month

AI Engineer

September 2023 - Present (2 years 1 month)

Bristol, England, United Kingdom

Explainable Reinforcement Learning (XRL) research for logistical management and mission planning using Proximal Policy Optimisation (PPO) and Deep Q-Learning (DQN).

- 2 academic papers accepted at international conferences (1 as first author).

Machine Learning Anomaly Detection and MLOps for Artemis Orion's European Service Module using Kubeflow Pipelines (Bremen, Germany).
Implementing algorithms such as isolation forest to analyse propulsion system data.

End-to-End Machine Learning Engineering (Bristol, UK):

- Airline delay and air quality prediction in a heavily regulated industry
- Exploratory data analysis with SQL
- Data engineering
- Model experimentation
- CI/CD pipeline implementation for model maintenance and serving
- REST API for online inference of airline delays and air quality

DevSecOps Software Engineering (Stevenage, UK):

- Implementing a CI pipeline for validation of software for the OneSat program

- Developing a CONOPS-style explanation for network interactions

Google Cloud Platform:

- Certified Professional Machine Learning Engineer
- Advanced Solutions Lab Advanced Generative AI Certification

Airbus Sloshing Rocket Workshop Finance Lead

September 2023 - August 2025 (2 years)

Bristol, England, United Kingdom

- Managing Airbus' sponsorship of the event to EUROAVIA
- Organising Airbus' representatives to attend the final competition in Lisbon, Portugal
- Procuring promotional material to provide to the students and volunteers of the event

Institution of Engineering and Technology (IET)

4 years 10 months

Artificial Intelligence Technical Network Committee Member

April 2025 - Present (6 months)

Bristol, England, United Kingdom

Supporting artificial intelligence education and information dissemination through events, webinars, podcasts and more!

Aerospace Technical Network Committee Member

January 2023 - Present (2 years 9 months)

Bristol, England, United Kingdom

Supporting the growth of the IET's Aerospace Technical Network which aims to bring together experts from aerospace-related fields and facilitate discussions about recent industry developments or collaborative ventures between those attending.

Authoring aerospace articles for publishing on IET's EngX platform.

IET South Australia & Northern Territory Local Network Committee Member

February 2021 - January 2023 (2 years)

Adelaide, South Australia, Australia

Representing the IET Adelaide University On Campus (AUOC) amongst the wider IET communities of South Australia and the Northern Territory.

Liaising with the South Australia and North Territory committee to discuss the performance of the IET AUOC, areas of collaboration, and methods to boost membership intake by hosting industry nights and attending careers fairs. Representing the IET Adelaide University On Campus (AUOC) amongst the wider IET communities of South Australia and the Northern Territory. Liaising with the South Australia and North Territory committee to discuss the performance of the IET AUOC, areas of collaboration, and methods to boost membership intake by hosting industry nights and attending careers fairs.

IET Adelaide University On Campus President
November 2021 - December 2022 (1 year 2 months)
Adelaide, South Australia, Australia

Strategically planning the expected path of the Club in accordance with the Club's values, goals, and the suggestions of other Committee members. Putting people first. Strategically planning the expected path of the Club in accordance with the Club's values, goals, and the suggestions of other Committee members. Putting people first.

IET Adelaide University On Campus Founder and Secretary
December 2020 - November 2021 (1 year)
Adelaide, South Australia, Australia

Ensuring that the club operates efficiently and ethically whilst supporting the smooth operation of the management committee. My roles include:

- Liaising with the President and Vice-President to organise committee meetings;
- Communicating with all committee members to plan agenda items;
- Creating and filing meeting minutes to record discussions, actions, results and future goals;
- Keeping an up-to-date record of important contact details for the committee and for specific events (e.g. safety officers and volunteers).

Lockheed Martin
Associate Software Engineer
February 2023 - August 2023 (7 months)
Adelaide, South Australia, Australia

Full-stack web application development:

- Angular, HTML, CSS, TypeScript/JavaScript
- WebSocket
- Python, Node.js
- DevSecOps

Exhibited at 2023's Science Alive! to showcase the possibilities that STEM fields hold for aspirational students.

Defence Science and Technology (DST)

2 years 3 months

Decision System Scientist

March 2021 - February 2023 (2 years)

Adelaide, South Australia, Australia

Decision Systems:

- Experimental research in distributed decision systems.
- Developing testbeds for distributed decision-making algorithms in complex environments with complex systems of agents.
- Architecting testbeds in a manner that streamlines the transferability of public research to that within the team.
- Onboarding DevOps practices to this development in collaboration with external organisations.

In conjunction with my Honours project for a Bachelor of Engineering (Mechanical and Aerospace), Unmanned Aerial Vehicle Physical Homing Relay System for Internet of Military Things Synergy (UAV PHORESIS). This Honours project involved the build of an augmented Tarot T810 hexacopter to allow Internet of Things (IoT) devices to be picked-up and dropped off. This technology aids in creating self-healing mesh networks of sensors for rapidly-changing physical spaces. The applications of this technology include, but are not limited to, bushfire monitoring, open-pit mine air sampling, and delivery of medical supplies.

Summer Intern

December 2020 - February 2021 (3 months)

Adelaide, South Australia, Australia

Adept Adelaide

Adept Rocketry Division Technical Team: Recovery

March 2021 - November 2021 (9 months)

Adelaide, South Australia, Australia

Ensuring the safe and controlled descent and landing of the rocket once it has achieved its apogee. After this, locating and collecting the rocket in a timely manner such that it could possibly be re-launched within 24 hours.

University of Adelaide
Undergraduate Research Assistant
July 2020 - October 2020 (4 months)
Adelaide, South Australia, Australia

Researching methods to minimise wind loads on heliostat arrays. Performing experiments on scaled heliostats of various sizes at Thebarton Wind Tunnel. Learning about academic research paradigms and effective experimental methods.

Glynde Mitre 10
Sales Assistant & E-Commerce Assistant
August 2016 - February 2020 (3 years 7 months)
Glynde, South Australia, Australia

Identifying and uploading items to the store's online product catalogue. Safely transport new stock to the shop floor. Photographing, pricing and describing products, then adding them to the online store. Designing and publishing specials and promotions. Managing cash, EFTPOS, and account payments. Correctly taking orders and managing refunds. Provide advice for garden-related questions. Assisting customers in transporting heavy or large products to their vehicles.

Saint Ignatius' College, Adelaide
Field Hockey Coach
2018 - March 2019 (1 year)
Athelstone, South Australia, Australia

Communicated with sports staff, parents, and players to organise teams appropriately for competition matches. Transferred my knowledge of field hockey to junior players. Dynamically adapted my teaching techniques to give player-specific advice.

Petro-King
Work Experience Student
January 2015 - February 2015 (2 months)
Adelaide, South Australia, Australia

Prioritising, sorting, and replying to internal emails regarding upcoming events/meetings. Searching the intranet and email inbox for upcoming meetings/events/conferences and organising a simple schedule without clashes. Learning about oilfield technology and searching for profitable oil deposits.

Education

EPFL (École polytechnique fédérale de Lausanne)

Space Mission Design and Operations (EE585x) · (May 2023 - July 2023)

University of Adelaide

Bachelor of Engineering - BE (Honours), Mechanical & Aerospace
Engineering · (2018 - 2022)

University of Adelaide

Bachelor of Mathematical & Computer Sciences - BMa&CompSc
(Honours), Mathematical · (2018 - 2022)

University of South Australia

Course (2-month), Astrophysics · (May 2023 - June 2023)

University of Southampton

Mechanical & Aerospace Engineering · (2019 - 2020)